Factors, Multiples, Squares and Cubes

Name _____ Date _____ No calculator Version 1 Scan to practise this topic online

KEY WORDS factor *(tauwehe)* · multiple *(tau whakarea)* · square number *(tau pūrua)* · cube number *(tau pūtoru)*

A Factors and factor pairs ●○○

Remember: A factor of a number divides into it exactly. Factors come in pairs: $12 = 1 \times 12 = 2 \times 6 = 3 \times 4$.

1. Write all the factor pairs of each number.

a) 12 _____ **b)** 18 _____

c) 20 _____ **d)** 30 _____

2. List all the factors of each number.

a) 8 _____ **b)** 15 _____

c) 24 _____ **d)** 36 _____

e) 13 _____ **f)** 40 _____

B Multiples ●○○

Watch out: Factors are the numbers that divide into a number. Multiples are the numbers in its times table. The factors of 6 are 1, 2, 3, 6. The multiples of 6 are 6, 12, 18, ...

3. Write the first five multiples of each number.

a) 3 _____ **b)** 7 _____

c) 9 _____ **d)** 12 _____

4. Circle the multiples in each list.

a) Multiples of 4: 6 12 18 20 30 36 42

b) Multiples of 6: 12 16 18 26 30 42 56

c) Multiples of 9: 18 24 27 39 45 54 63

C Square numbers ●●○

Remember: A square number is a number multiplied by itself: $7^2 = 7 \times 7 = 49$.

5. Work out.

a) 3^2 = _____ **b)** 6^2 = _____ **c)** 1^2 = _____ **d)** 9^2 = _____

e) 4^2 = _____ **f)** 11^2 = _____ **g)** 7^2 = _____ **h)** 12^2 = _____

i) 5^2 = _____ **j)** 10^2 = _____ **k)** 8^2 = _____ **l)** 2^2 = _____

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

6. Circle the square numbers.

a) Square numbers: 4 12 25 48 64 90 100 121

D Cube numbers ●●○

⚠ **Watch out:** 2^3 means $2 \times 2 \times 2 = 8$. It does not mean 2×3.

7. Work out.

a) 1^3 = _____ **b)** 2^3 = _____ **c)** 3^3 = _____ **d)** 4^3 = _____

e) 5^3 = _____ **f)** 10^3 = _____ **g)** $2^3 + 3^2$ = _____ **h)** $5^3 - 5^2$ = _____

8. Circle the cube numbers.

a) Cube numbers: 1 9 27 36 64 100 125 150

E Squared and cubed ●●○

💡 **Remember:** "5 squared" is written 5^2 and means 5×5. "4 cubed" is written 4^3 and means $4 \times 4 \times 4$.

9. Write each one using a small number (an exponent), then work it out.

a) 5 squared **b)** 4 cubed **c)** 9 squared

d) 2 cubed **e)** 12 squared **f)** 3 cubed

F Problems ●●●

10. A square garden has sides of 8 m. How many square metres of grass are needed to cover it?

11. A cube is built from small cubes. It is 3 cubes long, 3 cubes wide and 3 cubes high. How many small cubes are used?

12. 24 chairs are set out in equal rows. Write all the ways this can be done, as rows × chairs in each row.

★ Extension

13. Which number less than 100 is both a square number and a cube number, and is not 1?

14. Find a number that is a multiple of both 4 and 6. What is the smallest one?

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

Factors, Multiples, Squares and Cubes

Name _____ Date _____ No calculator Version 2 Scan to practise this topic online

KEY WORDS factor *(tauwehe)* · multiple *(tau whakarea)* · square number *(tau pūrua)* · cube number *(tau pūtoru)*

A Factors and factor pairs ●○○

Remember: A factor of a number divides into it exactly. Factors come in pairs: $12 = 1 \times 12 = 2 \times 6 = 3 \times 4$.

1. Write all the factor pairs of each number.

a) 10 _____ **b)** 16 _____

c) 28 _____ **d)** 24 _____

2. List all the factors of each number.

a) 6 _____ **b)** 21 _____

c) 32 _____ **d)** 48 _____

e) 11 _____ **f)** 45 _____

B Multiples ●○○

Watch out: Factors are the numbers that divide into a number. Multiples are the numbers in its times table. The factors of 6 are 1, 2, 3, 6. The multiples of 6 are 6, 12, 18, ...

3. Write the first five multiples of each number.

a) 4 _____ **b)** 6 _____

c) 8 _____ **d)** 11 _____

4. Circle the multiples in each list.

a) Multiples of 3: 9 14 18 22 27 31 36

b) Multiples of 7: 14 21 27 35 44 49 54

c) Multiples of 8: 16 20 24 36 40 56 62

C Square numbers ●●○

Remember: A square number is a number multiplied by itself: $7^2 = 7 \times 7 = 49$.

5. Work out.

a) 4^2 = _____ **b)** 7^2 = _____ **c)** 1^2 = _____ **d)** 8^2 = _____

e) 3^2 = _____ **f)** 12^2 = _____ **g)** 6^2 = _____ **h)** 11^2 = _____

i) 5^2 = _____ **j)** 10^2 = _____ **k)** 9^2 = _____ **l)** 2^2 = _____

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

6. Circle the square numbers.

a) Square numbers: 9 15 36 50 49 80 81 144

D Cube numbers ●●○

⚠ **Watch out:** 2^3 means $2 \times 2 \times 2 = 8$. It does not mean 2×3.

7. Work out.

a) 2^3 = _____ **b)** 1^3 = _____ **c)** 4^3 = _____ **d)** 3^3 = _____

e) 5^3 = _____ **f)** 10^3 = _____ **g)** $3^3 + 2^2$ = _____ **h)** $4^3 - 4^2$ = _____

8. Circle the cube numbers.

a) Cube numbers: 8 16 25 27 50 64 100 125

E Squared and cubed ●●○

💡 **Remember:** "5 squared" is written 5^2 and means 5×5. "4 cubed" is written 4^3 and means $4 \times 4 \times 4$.

9. Write each one using a small number (an exponent), then work it out.

a) 6 squared **b)** 5 cubed **c)** 8 squared

d) 3 cubed **e)** 11 squared **f)** 2 cubed

F Problems ●●●

10. A square garden has sides of 9 m. How many square metres of grass are needed to cover it?

11. A cube is built from small cubes. It is 4 cubes long, 4 cubes wide and 4 cubes high. How many small cubes are used?

12. 36 chairs are set out in equal rows. Write all the ways this can be done, as rows × chairs in each row.

★ Extension

13. Which number less than 100 is both a square number and a cube number, and is not 1?

14. Find a number that is a multiple of both 3 and 5. What is the smallest one?

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.